

Introduction

The objective of image restoration is to improve a given image in some predefined sense.

Image enhancement is a subjective process whereas image restoration is objective process.

Image restoration is to reconstruct or recover an image that has been degraded by using a prior knowledge of the degradation phenomena.

So, this technique is a process of degradation and applying the inverse process in order to recover the original image.

If there is a presence of noise and image blurring we use image restoration.

A model of Image Degradation / Restoration Process

The figure below shows a model of image degradation / restoration process:

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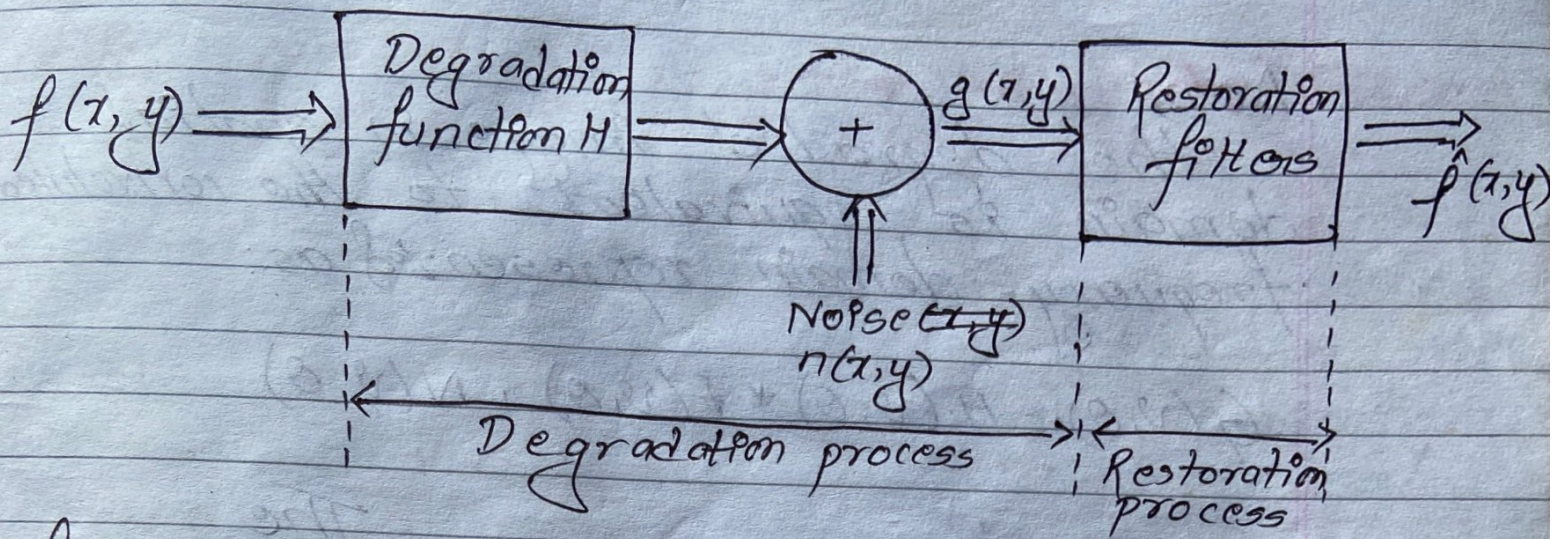


fig: A model of image degradation/restoration process—

The degraded image $g(x, y)$ can be produced as:

$$g(x, y) = H[f(x, y)] + n(x, y)$$

where,

$f(x, y)$ = I/p image

$n(x, y)$ = Additive noise term

Now, The degraded image can be defined in spatial domain as:

$$g(x, y) = h(x, y) * f(x, y) + n(x, y)$$

where,

$h(x, y)$ = Spatial representation of the degradation function

②

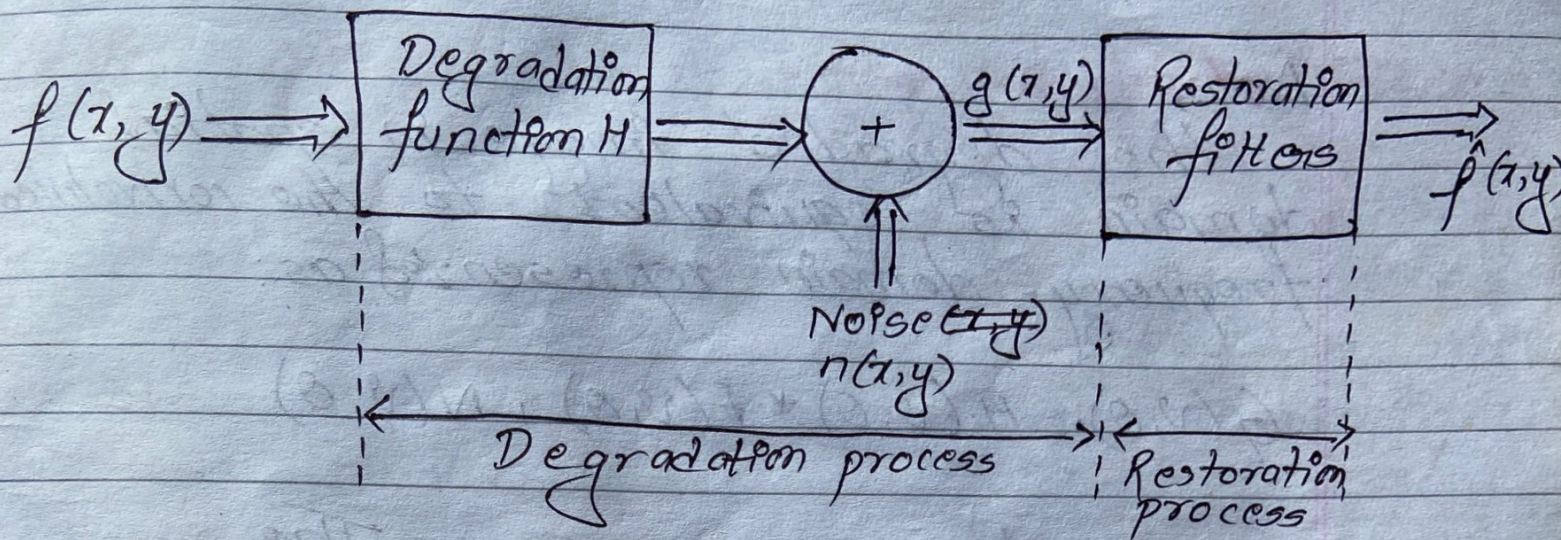


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Now,

the degraded image in spatial domain is equivalent to the convoluted frequency domain represented as:

$$G(v, \theta) = H(v, \theta) * F(v, \theta) + N(v, \theta)$$

The degradation process is also referred to as convolution of the image with a point spread function (PSF).

Similarly, restoration process is defined or referred to as deconvolution.

Hence,

Restoration process can be obtained as:

$$\hat{f}(x, y) = R[g(x, y)], \text{ In linear spatial invariant form}$$

$$\hat{f}(v, \theta) = R[G(v, \theta)], \text{ In deconvolution form.}$$

(4)

Noise Model Restoration in the Presence of Noise Only Spatial Filtering

The method of choice for reducing noise is spatial filtering. After filtering the noise, there should have a restoration process to have or obtain original image.

For noise reduction, following summarize and implement two types of spatial filters:

- (i) Spatial Noise filters
- (ii) Adaptive Spatial filters

Spatial Noise Filter

In spatial noise filters image independently of how image characteristics vary from one location to another location.

In this condition the spatial noise filters can be used & explained as:

(i) Arithmetic mean,

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(s, t) \in S_{xy}} g(s, t)$$

(i) Geometric mean,

$$\hat{f}(x, y) = \left[\prod_{(s, t) \in S_{xy}} g(s, t) \right]^{1/mn}$$

(ii) Harmonic mean,

$$\hat{f}(x, y) = \frac{mn}{\sum_{(s, t) \in S_{xy}} \frac{1}{g(s, t)}}$$

(iv) Median,

$$\hat{f}(x, y) = \text{Median} \{g(s, t) \mid (s, t) \in S_{xy}\}$$

(v) Max,

$$\hat{f}(x, y) = \max_{(s, t) \in S_{xy}} \{g(s, t)\}$$

(vi) Min,

$$\hat{f}(x, y) = \min_{(s, t) \in S_{xy}} \{g(s, t)\}$$

where,

m & n = variables denotes image rows & columns spanned by the filter respectively

S_{xy} denotes an $m \times n$ sub image of the input noisy image g .

$\hat{f}(x,y)$ = filter response at (x,y) condition (co-ordinates)

(x,y) = co-ordinates of sub image

2) Adaptive Spatial Filter

In some application, results can be improved by using filters capable of adapting their behaviour based on the characteristic of image in the region being filtered.

The adaptive median filtering algorithm uses two processing levels for adaptive spatial filter that can be denoted as:

Level A:

If $Z_{min} < Z_{med} < Z_{max}$ go to level B

Else, increase the window size

If window size $\leq S_{max}$, repeat level A

Else output Z_{med}

Level B:

If $Z_{min} < Z_{xy} < Z_{max}$, output Z_{xy}
 Else output Z_{med}

where,

Z_{min} = Minimum intensity value in S_{xy}

Z_{max} = Maximum intensity value in S_{xy}

Z_{med} = Mediating intensity value in S_{xy}

Z_{xy} = Intensity value at
 co-ordinate (x, y)

S_{xy} = maximum allowed size of the
 adaptive filter window.

Periodic Noise Reduction Using Frequency Domain Filtering

Periodic noise produces impulse like bursts that often are visible in the Fourier spectrum.

The principal approach for filtering these components is to use notch reject filtering.

The general expression for notch reject filtering having 'Q' notch pair can be defined as:

$$H_{NR}(u, v) = \prod_{k=1}^Q H_k(u, v)$$

$$H_{NR}(u, v) = \prod_{k=1}^Q H_k(u, v) H_{-k}(u, v) \quad \dots (i)$$

where,

$H_k(u, v)$ & $H_{-k}(u, v)$ are high pass filter with centre at (u_k, v_k) & $(-u_k, -v_k)$ respectively

These translated centres are specified with respect to the centre of the frequency rectangle $(M/2, N/2)$.

Therefore, the distance computation for the filters are given by following expression.

$$D_k(u, v) = \left[(u - M/2 - u_k)^2 + (v - N/2 - v_k)^2 \right]^{1/2}, \text{ for } (u_k, v_k)$$

and,

$$D_k(u, v) = \left[(u - M/2 + u_k)^2 + (v - N/2 + v_k)^2 \right]^{1/2}, \text{ for } (-u_k, -v_k)$$